

BARATH ROSHAN S

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SUMMARY

AI/ML Engineer with hands-on experience building computer vision, NLP, and full-stack ML systems using Python, TensorFlow, FastAPI, and modern deep learning frameworks. Proven track record across internships, hackathons, and independent projects, including real-time detection systems, document intelligence pipelines, and production-style AI applications integrating cloud and database services.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Data Structures and Algorithms, OOPS.
AI/ML & Frameworks: TensorFlow, Scikit-learn, PyTorch, Hugging Face Transformers, OpenCV, FastAPI, Django, Neural Networks, Embedding Models, MCP, RNN, LSTM, API Integration, EDA.
Web & Databases: React.js, Node.js, MySQL, MongoDB, PostgreSQL, SQLite
Tools: Git, GitHub, VS Code, IntelliJ IDEA, Google Colab, Power BI, Roboflow, Canva

EXPERIENCE

- AI & Data Analytics Intern — NIELIT, Calicut

Dec 2025

 - Built AI-based fire detection and violence detection systems for real-time surveillance using Python, OpenCV, TensorFlow, and YOLO.
 - Managed end-to-end pipeline including data collection, preprocessing, model training, fine-tuning, and performance evaluation to improve detection accuracy and reliability.
 - Optimized and integrated trained models into practical workflows, improving operational efficiency and robustness.
- AI/ML Intern — Pinesphere Solutions Pvt Ltd

May–Jun 2026

 - Developed and fine-tuned deep learning models for intelligent data extraction and automation using Python and modern AI frameworks.
 - Built scalable backend APIs and integrated databases and cloud services to support efficient AI-driven data processing.
 - Designed end-to-end full-stack applications following software engineering best practices for production-ready delivery.

PROJECTS

- Rakshitatha — AI-Powered Parametric Insurance System

 - Built a data-driven disruption and payout engine for gig-worker protection using FastAPI and machine learning to predict delivery delays.
 - Detected real-time environmental disruptions and automated dynamic compensation using Weather, AQI, and Geolocation APIs.
 - Implemented K-Means-based risk zoning across operational zones to support smarter decision-making.

Tech: Python, FastAPI, Scikit-learn, Pandas, NumPy, Realtime APIs.
- PropVision AI — AI-Powered Property Discovery Platform

 - Developed an AI-powered property discovery platform trained on 14,000+ real estate images using deep learning and computer vision.
 - Implemented image super-resolution, room classification, and semantic search to enhance property exploration and decision-making.

Tech: Python, FastAPI, React.js, Tailwind CSS, Real-ESRGAN, EfficientNet-B4, CLIP, PostgreSQL.
- InvoiceIQ — AI-Powered Invoice Document Intelligence System

 - Built an invoice processing platform by fine-tuning a Donut Document Understanding Transformer on invoice datasets.
 - Enabled intelligent field extraction, document validation, and structured data generation from scanned invoices and PDFs.

Tech: Python, FastAPI, Donut Transformer, PyTorch, Hugging Face Transformers, OpenCV, MongoDB.

ACHIEVEMENTS

- Top 100 Team, Guidewire DEVTrails University Hackathon 2026 — contributed as ML Engineer building predictive models for an intelligent insurance automation system.
- Top 10 Position, Datathon 2026 (Coimbatore Institute of Technology) — developed ML workflows for image enhancement, room classification, semantic property search, and AI-agent query understanding.

EDUCATION

B.Tech, Artificial Intelligence & DataScience—SriEshwarCollegeofEngineering	2024 – 2028
CGPA: 7.86 (up to 3rd semester)	
HSC— Bharani ParkMatric Hr SecSchool	2022 – 2024
93.3%	
SSLC— Cheran Matriculation School	2021 – 2022
92.8%	

CERTIFICATIONS & CODING PROFILES

- Certifications: B2-Linguaskills Business, Elements of AI (University of Helsinki), Generative AI (NVIDIA), Python (Coursera)
- CodeChef: 600+ problems solved, Max Rating 1235 | LeetCode: 250+ problems solved, Max Rating 1489
- HackerRank: 3-Star in Python | SkillRack: 700+ problems solved, 160+ Bronze badges